Deficiency

As with the other two electrolytes, chloride deficiency rarely results from poor dietary intake. Rather, chloride deficiencies are typically caused by excessive losses due to diarrhea or vomiting, heavy perspiration, trauma, or kidney disease.

Micro minerals

We require intakes of the essential trace minerals, or micro minerals, in amounts of less than 100 milligrams per day. The essential trace minerals are iron, zinc, copper, iodine, manganese, fluoride, chromium, selenium, and molybdenum.

Iron

About 30% of the iron in the body is stored in the spleen, liver, and bone marrow. Iron is a constituent of haemoglobin and myoglobin, which are iron-binding proteins found in red blood cells and muscle, respectively. Iron also function as a cofactor for a number of enzymes and is required for their activity.

Deficiency

Iron deficiency develops in stages; the last stage is iron-deficiency anemia.

Zinc

Most zinc in the body is found in the bones and muscles. This trace mineral is necessary for optimal activity of many enzymes and for various bodily processes, including the following:

- Protein metabolism, wound healing, and growth
- Metabolism of DNA, the genetic material
- Development of sexual organs and bones
- Immune responses
- Memory formation
- Alcohol metabolism

Particularly high needs for zinc occur during growth and development.

Food sources

Meat, fish, eggs, milk and nuts are rich sources.

Deficiency

A zinc deficiency impairs protein synthesis, collagen formation, and energy production; it also decreases alcohol tolerance.

Copper

The trace mineral copper is required for proper use of iron by the body and for the activity of certain enzymes. This trace mineral also aids in the production of connective tissue, blood vessels, phospholipids, and melanin (a skin pigment).

Food sources

The main dietary sources of copper are shellfish, whole grains, legumes, and nuts.

Deficiency

Symptoms include decreased blood levels of copper, fewer white blood cells, anemia, bone demineralization, deterioration of the nervous system.

Iodine

The micro mineral iodine is a necessary component of two hormones produced by the thyroid gland (thyroxine and triiodothyronine). These thyroid hormones are critical in regulating the body's metabolic rate.

Food sources

The amount of iodine present in a particular food depends on the amount of iodine in the environment in which that food was raised. Therefore, food from the sea lobsters, oysters, sardines etc.

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